

Scalability & Future Plan

Date	05 June 2026
Team ID	SWTID-2026-2950
Project Name	Finrelief (AI Powered Debt Relief & Financial Recovery Platform)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	SQLite Single-Write Lock	The SQLite database blocks concurrent operations during write transactions. Under high user loads or synchronous writes, users experience database-locked errors and request timeout failures	High
2	Synchronous Gemini API Operations	Triggering direct, blocking calls to the Gemini API for every advisor interaction slows request latency to 2–5 seconds and makes the application sensitive to quota overruns.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-node backend. Async FastAPI handles network load, but performance degrades sharply on database write contention.	Containerize using Docker and scale horizontally across multiple stateless nodes behind an Nginx load balancer.
2	Data Storage	Relational data is saved inside a local SQLite database with no backing volume. Documents are saved as raw text.	Migrate transactional tables to PostgreSQL (Supabase/RDS). Integrate AWS S3 or Google Cloud Storage for hardship letter document storage.
3	Performance	Synchronous database calls and real-time AI responses execute blocking routines. No API caching is configured.	Integrate Redis for session and schema caching. Offload heavy AI text generation processes to background Celery task queues.
4	Security	Auth JWT keys stored in browser `localStorage`. Basic CORS setup. No active IP or API endpoint rate limiters	Store authentication tokens in HTTP-Only secure cookies. Deploy the backend behind a WAF and apply route rate-limiting via `slowapi`.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Production Infrastructure & Security Hardening	Month 1–2	• Persistent, high-availability PostgreSQL backend. • Secure token cookies preventing XSS token theft
Phase 3	Semantic Caching & PDF Document Generation	Month 3–4	• Reduced Gemini API costs (up to 60%) using pgvector RAG caching. • Clean, print-ready PDF downloads for creditor hardship letters.
Phase 4	Plaid API Sync & Alert Crons	Month 5–6	• Automated bank balance and debt APR synchronizations. • Recurring notifications alerting users to upcoming payoff deadlines.